

HW

1) Java generates Hashcode for every object.

Findout what is the hash code in java & the purpose of generating the Hashcode

A hash code is a unique integer value generated for an object using the hashCode() method of the Object class. It helps Java identify and organize objects efficiently. Different objects may sometimes have the same hash code but objects that are equal must have the same hash code.

Purpose of Generating a Hashcode:

1. Fast searching and retrieval of objects
2. Improves performance by reducing time
3. Supports object comparison
4. Efficient storage.

Ex. class Demo

```
public static void main(String[] args)
{
    String s = "Java";
    System.out.println("Hashcode: " + s.hashCode());
}
```

Output:

Hashcode: 2301506

Q1/7

HW

Q) What are the various problems which you have observed in community & Society. Pick any 5 problem find out if they can be solved using Data Structures and Algorithms.

1. Traffic Congestion

Problem: Heavy traffic causes delay and fuel wastage.

Solution using DSA: Use Graphs and Shortest Path Algorithms to find the fastest route.

2. Hospital Appointment Management:

Problem: Long waiting time for patients.

Solution using DSA: Use Graphs & Rout.

Use Queue & Priority Queue to schedule patients efficiently.

3. Waste collection Optimization

Problem: Garbage is not collected efficiently.

Solution using DSA: Use Graphs & Route Optimization Algorithms to plan the shortest collection routes.

4. Fake certificate verification:

Problem: Educational Institutes receive fake certificates.

Solution using DSA: Use Hash Tables & Searching Algorithms to quickly verify certificate records.

5. Library book management:

Problem: Finding and issuing books takes time.

Solution using DSA: Use Binary Search Trees & Hash tables for fast searching and book management.